Lab #3 : Data preprocessing

Machine Learning *Academic Year : 2026/2027*
Prof. SAYAD Lamri *Duration : 1h30*

Software requirements: Python 3, Anaconda, Jupyter Notebook
Objectives :

- Inspect and understand a dataset.
- Identify numerical and categorical variables.
- Detect and handle missing values.

Step 1 : Dataset inspection

1. Load the "titanic.csv" dataset into a dataframe "data1"
2. Determine the dimensions of this dataset
    a. How many observations
    b. How many features
3. Inspect the data types
    a. Identify numerical features
    b. Identify categorical features
    c. Identify the target variable
4. Display statistic summary
    a. Display the number of missing per feature
    b. Display the percentage of missing per feature
    c. Which columns contain missing values
    d. Should we remove the entire column
    e. Should we remove rows
    f. Could we impute the missing values

Step 2 : Remove unnecessary features

5. Remove unnecessary features :
    a. Remove the feature : "PassengerId"
    b. Remove the feature : "Name"
    c. Remove the feature : "Ticket"

Step 3 : Identify missing values

6. Display the number of missing values per feature
7. Display the percentage of missing values per feature
8. Questions :
    a. Which columns contain missing values
    b. Should we remove the entire column
    c. Should we remove rows
    d. Could we impute the missing values

Step 4 : Handling missing values

9. Remove rows :
    a. Create "data2" a copy of the dataframe "data1"
    b. Display the dimensions of "data2"
    c. Remove all rows with missing values from "data2"
    d. Display the dimensions of "data2"
    e. What is the disadvantage of this approach
10. Mean Imputation :

a. Create “data3” a copy of the dataframe “data1”
b. Display the number missing values in the column “Age”
c. Apply mean imputation on “data3” to the variable “Age” to fill missing values with the mean value.
d. Display the number missing values in the column “Age”
e. What is the disadvantage of this approach

11. Median imputation
    a. Repeat steps e, f, g and h by applying median imputation instead of mean imputation
    b. Why might the median be preferable to the mean
12. Categorical imputation
    a. Apply categorical imputation to the categorical feature “Embarked” (use mode)

Step 5 : Handling duplicate observations

13. Create “data4” a copy of the dataframe “data1”
14. Display the dimensions of “data4”
15. Display the number of duplicate rows
16. Remove duplicate rows from “data4”
17. Display the dimensions of “data4”
18. Questions :
    b. Why can duplicates be problematic?
    c. Should duplicates always be removed?
    d. What if two people legitimately have identical feature values?

Step 6 : Encoding categorical variables

19. Transform the column “Sex” by applying binary encoding
20. Which encoding approach is more appropriate to transform the column “Embarked”
    a. Apply this approach to transform the column Embarked
    b. Display the dimensions of the dataframe

Step 7 : Feature scaling

21. Scale all features by applying Min-Max scaling
22. Scale all features by applying standardization